Abhinav Danthuri

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Summary

Computer Science graduate (AI/ML specialization) who builds **agentic AI systems** using Python, LangGraph, RAG, and FastAPI. Strong fundamentals in Java, OOP, and data structures. Seeking an entry-level **AI Developer** role where I can ship production-quality AI features.

Education

ACE Engineering College, Hyderabad **2022 – 2026**
Bachelor of Technology in Computer Science Engineering *7.65 / 10 CGPA*

Projects

Autonomous Bug Investigator **Jun 2026 – Sep 2026**
Python, LangGraph, FastAPI, FAISS, MySQL, GitHub API, React.js

- Built a tool that takes a GitHub issue and searches the repo's **code, docs, past issues, and commits** to find the likely root cause.
- Used **LangGraph** to split the work across planner, investigator, fix, and critic agents. The critic sends weak answers back for another pass.
- Generates a debugging report, a suggested fix, and test cases, and saves every run's history in **MySQL**.

SILENT: Agentic AI-Based Silent Speech Recognition System **Oct 2025 – Dec 2025**
Python, OpenCV, PyTorch, CNN, LLM, RAG, LangGraph, FAISS, FastAPI

- Built a lip-reading system that works out speech from video alone, using a **PyTorch CNN** with an **LLM** to clean up its predictions.
- Set up a **LangGraph** agent pipeline with **FAISS**-based **RAG** that steps in when the CNN is unsure and corrects it using context.
- Wrapped it in a **FastAPI** backend with a simple UI showing predictions, confidence, and the final sentence.

RL-Based Smart Scaling for SLA-Adherent Cloud Resource Management **Jan 2026 – Mar 2026**
Python, Reinforcement Learning, Q-Learning

- Analyzed cloud workload patterns to identify performance gaps and limitations of static autoscaling approaches.
- Modeled cloud resource allocation as a **Reinforcement Learning problem using Python** and trained a **Q-Learning** agent to automate scaling decisions.
- Benchmarked against static autoscaling, achieving a **28% reduction in SLA violations** and **19% lower resource consumption** across varying cloud workloads.

Technical Skills

AI & Machine Learning:PyTorch, CNN, RL, LLMs, RAG, LangGraph, FAISS,OpenAI API, Prompt Engineering
Languages: Python, Java, JavaScript, SQL
Core CS: OOP, Data Structures & Algorithms
Frameworks & Data: FastAPI, React.js, MySQL
Tools & Platforms: Git, GitHub, Docker, Jupyter Notebook, VS Code, IntelliJ IDEA

Certifications

- Advanced Java Programming and Data Structures
- The Ultimate MySQL Bootcamp
- JavaScript Essential Training
- Pearson MePro English Proficiency – Level 10